LPS / PS-LPS Program: Plan of Action and Phase Sequencing

v2 — refined as a base × refinement algebra

A dependency-ordered roadmap: two base estimators (LPS, local covariance), the prediction-synchronization and local dimension/scale refinement axes, the anchoring choice, the simplicial-complex integration, and the occupancy capstone

Internal program plan — LPS / `geosmooth` program

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Abstract

The first plan listed the program as a linear sequence of phases. Working through the local-chart geometry made a cleaner structure visible: the program is a small *algebra*. There are two **base estimators** — LPS (the local-polynomial conditional mean) and LCov (the local conditional covariance/association) — and a short list of **refinement operators** that apply to either base and *compose*. The operators fall on two orthogonal axes: **(A) prediction synchronization** (PS), which couples overlapping local models across anchors, and **(B) the local dimension/scale ladder**, which sets each anchor's local chart. Axis B has four rungs of increasing structure — fixed dimension/scale (the base); **LDS**, the joint per-anchor estimation of *both* local dimension and scale, unregularized; **LDR**, regularization of the dimension field with scale held fixed (the original framing); and **LDSR**, co-regularization of dimension *and* scale together. A third, cross-cutting axis is the **anchoring** choice: local models centered at the data points, or at a regular/derived grid. The two refinement axes are genuinely orthogonal — and they *meet* at the chart nerve, which is therefore named as the integration phase rather than a sub-part of any one operator. The validated base emits the reusable S/df/LOO/band toolkit; the synthetic-data enabler runs throughout; everything converges into the nerve integration and then the occupancy / quasi-equilibrium capstone and the terminal 16S application. The whole program runs under the two-agent audit-independence workflow. As of this revision, Tier-0 (the LPS base) is validated and merged, and the Tier 1–4 estimator gates are in integration.

Contents

1 How to read this, and the one rule that wraps every phase

This is a sequencing document, not a methods document; the methods live in the design notes referenced at the end, and each phase points to its note. The job here is to say what is built first, in what combinations, and why the order is forced by dependencies rather than preference.

One rule wraps every phase: each is executed under the *two-agent audit-independence workflow* — an implementer builds; an independent auditor validates against a fixed charter (the auditor's scope is not set by the implementer; the handoff carries evidence and admitted limitations but not the audit's questions; "report rendered" is never "phase accepted"; and a gate is real only if it can be made to fail). A phase is not complete — and dependents do not start — until its go/no-go gate is passed by that independent audit or adjudicated by the investigator.

2 The map: two bases, two refinement axes, one anchoring choice

Two base estimators. **LPS** estimates the conditional mean $\mathbb{E}[y \mid x]$ by local polynomial regression in a local chart. **LCov** estimates the local conditional second-order structure (covariance / precision / association) in the same chart geometry. They share the chart machinery; they differ in what is fit on the chart.

Axis A — prediction synchronization (PS). An operator that couples *overlapping* local models: where two anchors' charts share data, their predictions are regularized toward agreement (prediction-overlap coupling, strength λ_{sync}). PS acts *across* anchors, on the overlap graph. It applies to LPS (the existing PS-LPS) and, by analogy, to LCov.

Axis B — the local dimension/scale ladder. An operator that sets *each* anchor's local chart — its intrinsic dimension d_i and its scale K_i (support size / bandwidth). Four rungs:

base
: d and K fixed (a numeric `chart.dim` and one CV-selected global K): the current default.

LDS — local dimension *and* scale, unregularized
: Estimate *both* fields per anchor, jointly. This rung is itself two coupled directions — dimension and scale — because the scale at which a neighborhood is read determines its apparent dimension and vice versa. No spatial smoothing: the honest raw estimator.

LDR — local dimension regularization

Regularize the dimension field (ℓ_1/TV over a graph) while holding scale fixed/global. This is the program's *original* way of thinking about local dimension; it is the special case of the next rung with scale-regularization switched off.

LDSR — local dimension *and* scale regularization

Co-regularize both fields together over the chart nerve, subject to the points-per-parameter coupling. The full object; it nests LDR.

Axis C (cross-cutting) — anchoring. Where the local models are centered: at the **data points** (the current choice), or at a **regular/derived grid** that decouples the anchor set from the sample (§8). This choice crosses every cell of the base×refinement product.

The product. The named variants are the cells of $\{\text{LPS}, \text{LCov}\}\times\{$base, PS, LDS, LDR, LDSR, PS+LDSR$\}$, each under a choice of anchoring:

Refinement	base = LPS	base = LCov
none	LPS	LCov
+ PS (synchronization)	LPS+PS (= PS-LPS)	LCov+PS
+ LDS (local dim & scale, raw)	LPS+LDS	LCov+LDS
+ LDR (dim regularization, scale fixed)	LPS+LDR	LCov+LDR
+ LDSR (dim & scale co-regularization)	LPS+LDSR	LCov+LDSR
+ PS + LDSR (full)	LPS+PS+LDSR	LCov+PS+LDSR

Why this ordering The two refinement axes are *orthogonal*: PS chooses how anchors talk to each other; the dimension/scale ladder chooses what each anchor's chart is. That is why they compose into LPS+PS+LDSR rather than competing — and why both ultimately live on the same object, the nerve of the chart cover (§10): PS's overlap graph and LDSR's co-regularization graph are the *same* graph. The nerve is therefore the integration phase, not a sub-part of either operator.

3 Phase 0 — the LPS base (the foundation that emits the tools)

Objective. Prove the base smoother is what it claims, and produce the reusable estimator toolkit every refinement reuses. **Inputs.** None (the foundation). **Deliverables.** The Tier-0 correctness battery (polynomial reproduction; the linear-smoother identity $\hat{y} = Sy$ with $\operatorname{tr} S$ and analytic LOO/GCV; boundary behaviour; consistency; binary recovery and calibration; CV no-leakage; degenerate geometry), then the Tier 1–4 plan (honest nested/grouped CV; bandwidth and kernel semantics; binary-mode hygiene; uncertainty bands). Spec: the experimental plan note. **Gate.**

Go/no-go gate The Tier-0 battery passes deterministically and the linear-smoother toolkit (S, df, LOO/GCV, bands) is implemented and tested. This is both the correctness gate and the *source* of the tools the refinements require: df-matched comparisons (PS), per-anchor local GCV (the dimension/scale ladder), and bands (LCov, capstone).

Status (this revision). *Met.* Tier-0 is validated and merged to `main`; the S/df/LOO/band toolkit exists (`R/lps_uncertainty.R`). Tier 1–4 gates (bandwidth multiplier; nested/grouped CV; binary hygiene and the ridge-alignment and NA-consistency fixes; pointwise bands) are in integration.

4 Axis B — the local dimension/scale ladder

This axis is the heart of the refinement. It is developed bottom-up: build the raw joint estimator (LDS) first, then the two regularizations (LDR, LDSR). Spec: the implementation notes (dimension) and the per-point-scale sketch.

4.1 LDS — joint local dimension and scale (the estimator)

Objective. Estimate, per anchor, *both* the intrinsic dimension d_i and the scale K_i (support / bandwidth), recognizing that they are coupled: the locally MSE-optimal scale depends on local curvature, density, noise, and d_i, and d_i is read off the neighborhood spectrum at scale K_i. **Inputs.** Phase 0 (the chart machinery and the S_{ii}/LOO toolkit); the toy synthetic generators. **Deliverables.** The joint per-anchor (d_i, K_i) fields. *One design fork decides whether the Phase-0 toolkit survives unchanged:* an **X-only** scale (geometry/density-driven; keeps $\hat{y} = Sy$ linear and the df/LOO/band toolkit exact) versus a **y-aware** scale (local GCV/LOO or Lepski; better MSE but S becomes y-dependent, so df/bands move to the selection-aware path). The honest caveat: response-removal LOO $\neq$ training-removal LOO for a k-NN bandwidth, so the cheap S_{ii} shortcut is a biased proxy for *scale* selection (favour Lepski, or a leave-a-block-out, for the scale field). **Gate.**

> **Go/no-go gate** On a known varying-scale / varying-dimension DGP, the raw (d_i, K_i) fields track the truth (smaller K in finer regions; d_i matching the stratum dimension), and the X-only variant preserves the $\hat{y} = Sy$ identity exactly. The fields will be high-variance — that is expected and is what the next two rungs address.

Unblocks. LDR, LDSR.

4.2 LDR — local dimension regularization (scale fixed)

Objective. Stabilize the per-anchor dimension field by ℓ_1/total-variation regularization over a graph, holding scale at the global value. (The program's original local-dimension picture.) **Inputs.** Phase 0; LDS's raw dimension field; the toy generators. **Deliverables.** The exact graph-cut/TV solve; the test battery T1–T9 (graph-cut exactness, ℓ_1-keeps-the-boundary vs ℓ_2-blurs, denoising, the boundary safeguard, scale/persistence, the coordinate trap, the support cap and soft face, the frontier, downstream Truth-RMSE). Spec: the implementation notes. **Gate.**

> **Go/no-go gate** The implementation notes' go/no-go: *(the nominal– effective dimension gap is large)* $\wedge$ *(the field improves Truth-RMSE on the stratum-mixture generator)*, without blurring a true dimension boundary. If either fails, the within-face field is not promoted; the cheaper soft-face-and-cap refinement is kept.

Unblocks. The nerve integration; it is also the scale-frozen baseline LDSR must beat.

4.3 LDSR — co-regularized dimension and scale

Objective. Co-regularize the (d, K) fields jointly over the chart nerve, subject to the points-per-parameter coupling, so scale adapts where the data's natural scale changes and the two fields stay spatially coherent. **Inputs.** LDS (the raw fields); LDR (the dimension-field solver and its ℓ_1-

boundary property); the chart nerve (§10). **Deliverables.** The coupled objective $\sum_i \ell_i(K_i, d_i) + \lambda_K \operatorname{TV}_G(K) + \lambda_d \operatorname{TV}_G(d)$ subject to $\binom{d_i+p}{p} \le K_i$, solved on the nerve graph G with the LDR machinery; identifiability guards (the points-per-parameter constraint; a parsimony tie-break; never selecting scale on in-sample RSS). Spec: the per-point-scale sketch. **Gate.**

> **Go/no-go gate** LDSR beats LDR (scale-frozen) at *matched df* on heteroscale DGPs, and reduces to $\approx$ the global K on a homogeneous truth (no spurious variation); the ℓ_1 co-regularization preserves a true joint scale/dimension boundary that ℓ_2 would smooth away.

Unblocks. LPS+PS+LDSR; the LCov ladder; the capstone's basin geometry.

5 Axis A — prediction synchronization (PS / PS-LPS)

Objective. Validate, and put on firm footing, the synchronization of overlapping local models through prediction-overlap regularization. **Inputs.** Phase 0 (the validated base *and* the toolkit). PS is independent of the dimension/scale ladder — it can be developed in parallel. **Deliverables.** Exploit PS-LPS's linearity in the response: exact effective df = tr $S(\lambda_{\text{sync}})$ and analytic LOO/GCV in place of K-fold CV; compare to LPS at *matched df* (not matched λ); fix the ridge-path discontinuity at $\lambda_{\text{sync}} = 0$; make the synchronization graph and weights explicit; characterize the estimand and the $\lambda_{\text{sync}} \to \infty$ limit; add bands. Spec: the audit memo's PS-LPS sections and the chart-aware synchronization. **Gate.**

> **Go/no-go gate** On synthetic data, PS improves over a *df-matched* LPS where the geometry predicts; λ_{sync} selection is well-founded via tr S (no grid-boundary railing); $\lambda_{\text{sync}} = 0$ reproduces LPS exactly and the path is continuous.

Unblocks. LPS+PS+LDSR; the nerve integration.

6 Composition — LPS+PS+LDSR

The two axes compose because they are orthogonal: LDSR sets each anchor's chart; PS couples the anchors' predictions. The natural order is to fix the chart field first (LDS→LDR/LDSR), then synchronize on it (PS), because PS's overlap graph is defined by chart overlaps, which the scale field determines. The fully composed object LPS+PS+LDSR is exactly the cochain picture of the nerve substrate: one consistency objective over the overlaps combining prediction agreement (PS) and a coherent chart-geometry field (LDSR). It is built in §10, not before, because it depends on all of its parts.

7 The local covariance base (LCov) and its parallel expansions

Objective. Build and validate the local conditional covariance / precision / association on the boundary-safe compositional (L^∞ / Aitchison) geometry — the second base estimator — and carry the *same* refinement algebra onto it. **Inputs.** Phase 0; the realistic 16S generators; a boundary-aware compositional front end (structural-zero and soft-face handling) that this base must establish first. **Deliverables.** Recovery of a known local covariance on synthetic 16S; the twolayer association object (co-occurrence support + conditional-abundance precision) with soft-face and ℓ_1-fusion. Then the parallel expansions: **LCov+PS** (synchronize overlapping local covariances — the second-order analogue of PS-LPS), **LCov+LDS/LDR/LDSR** (the *same* chart-geometry ladder; the dimension/scale machinery is shared with LPS), and **LCov+PS+LDSR**. Spec: the chart-aware association design and its addendum. **Gate.**

> **Go/no-go gate** The covariance/precision is recovered on synthetic data with known structure; the boundary-safe path never reintroduces a pseudocount; the ℓ_1 fusion preserves a planted reorganization across a community-state boundary that ℓ_2 would smooth away. LCov+LDSR reuses the validated LPS ladder, so its gate is the recovery gate plus the ladder's reduction/-tracking checks.

Unblocks. The nerve integration; the capstone.

> **Why this ordering** The chart-geometry ladder (Axis B) is genuinely *shared* between LPS and LCov: both fit something on a local chart whose dimension and scale must be chosen, so the same (d_i, K_i) field machinery serves both. Only the synchronization analogue (PS for second-order objects) and the compositional front end are new to LCov.

8 Axis C (cross-cutting) — grid anchoring

Objective. Decouple the *anchor set* (where local models are centered) from the *sample*: instead of one local model per data point, center them at a chosen set of anchors — a regular or derived grid. **Why it matters.** The estimand becomes a function on a controlled anchor set (resolution chosen, not dictated by sampling density); charts are reusable (predictable cost); the synchronization graph and the nerve are built on the anchors, so the anchor choice is upstream of PS and the nerve, not a detail. **Ways to define the grid.** A naive ambient lattice is wrong (anchors land in empty regions); useful constructions all keep anchors *on the data manifold*: farthest-point / Poisson-disk sampling of the data; a lattice on the estimated manifold (intrinsic coordinates); coarse-to-fine adaptive refinement driven by the local-scale field K_i (denser anchors where scale is fine); or the nerve vertices themselves. The local-scale field of LDSR is the natural driver of an adaptive grid — closing a loop between Axis B and Axis C. **Gate.**

> **Go/no-go gate** On synthetic data, grid-anchored LPS matches data-anchored LPS in accuracy at equal effective resolution, with anchors provably covering the data support (no anchor with an empty or degenerate neighborhood); an adaptive grid driven by K_i uses fewer anchors at equal accuracy than a uniform grid.

Unblocks. A cleaner nerve (a controlled cover) and predictable cost for the capstone on large real data.

9 Cross-cutting — the synthetic-data and VALENCIA enabler

Objective. Supply known-ground-truth, increasingly realistic non-manifold datasets, because every refinement's gate needs an answer key the real data cannot give. **Deliverables.** Early: the toy stratified generators (the DGP library, now frozen as Amendment 1) needed by LDS/LDR.

Adding for this revision: a **varying-scale** generator (a localized region of finer structure / higher curvature) for the scale ladder, alongside the varying-dimension (G4) and curved (G3) generators. Maturing: the VALENCIA E1/E2/E3 exploration producing realistic 16S generators for LCov and the capstone. Spec: the implementation notes. **Gate.**

> **Go/no-go gate** Toy and varying-scale generators ready before the ladder's gates; realistic generators pass a posterior-predictive fidelity check before any LCov/capstone result built on them is trusted.

> **Why this ordering** This workstream leads and runs throughout: it is a *dependency* of the ladder, LCov, and the capstone, so it cannot trail them. The Tier-0 DGP library (Amendment 1, G1–G7) is the first installment and is already audited and merged.

10 Integration — the simplicial complex (nerve)

Objective. Unify, on the nerve of the chart cover, the pieces that the product structure keeps separate: synchronization (PS), the dimension/scale field (LDSR), and the covariance (LCov), as one consistency objective over the overlaps. **Inputs.** The PS axis, the LDSR rung, the LCov base; the anchoring choice that defines the cover. **Deliverables.** Minimal-first: replace PS's pairwise overlap graph with the nerve 1-skeleton of variable-size charts; add 2-simplices (higher-order synchronization at junctions); co-regularize chart size and dimension over the nerve (this *is* LDSR's graph G); and, if it pays, the single cochain objective combining prediction, covariance, and dimension/scale. Spec: the chart-nerve substrate note. **Gate.**

> **Go/no-go gate** On synthetic data, the nerve 1-skeleton of variable-size charts beats the fixed k-NN overlap graph on held-out accuracy; 2-simplex synchronization either measurably helps at junctions or is dropped; the integrated objective is validated on known truth.

Unblocks. Phase 3 — it supplies the basin-and-corridor skeleton.

11 Capstone — occupancy / quasi-equilibrium / support

Objective. Find the pure components (recurrent ecological basins) and the transition phases between them. **Inputs.** LCov, the nerve / basin-corridor skeleton, the realistic 16S generators, and the COT occupation-measure machinery. **Deliverables.** Level-set basins on the chart cover; the geometric corridor discriminator (overlap mass + elongated covariance); the local-covariance basin shape; alignment of per-subject basins onto shared components; and the transfer operator on the coarse basin-and-corridor skeleton that earns the "quasi-equilibrium" label. Spec: the COT bridge note. **Gate.**

> **Go/no-go gate** On synthetic 16S with *known* components: geometry-first level-set basins are more stable across scales/bootstraps than the dictionary-factorization atoms; the transfer-operator metastable count agrees with the cluster-tree count; the corridor discriminator removes spurious components. Then, and only then, the HMP/152-subject experiment.

Unblocks. The terminal application.

12 Terminal — the real 16S / omics application

Objective. Estimate conditional expectations of binary and continuous features, and the community occupancy/component structure, on real 16S and other omics data, with honest uncertainty. **Deliverables.** Real-data conditional-expectation surfaces with bands; occupancy/component decompositions; reported under nested and grouped (subject-level) cross-validation so the headline numbers are not selection-optimistic or leakage-inflated. **Gate.**

> **Go/no-go gate** Held-out predictive validity on real data and biological interpretability of the components — judged, as throughout, by prediction, not by intrinsic-geometry scores.

13 Dependencies and decision gates that can re-route the program

The forced order, stated as dependencies:

1. **LPS base (Phase 0)** first — it emits the S/df/LOO/band toolkit the refinements reuse. *(Met.)*
2. **The two refinement axes develop in parallel on the base.** PS (§5) and the dimension/scale ladder (§4) are independent; within the ladder, **LDS before LDR/LDSR** (regularize an estimator you have).
3. **LCov base** after Phase 0; it *reuses* the chart-geometry ladder, so LCov's ladder rungs follow the LPS ones cheaply.
4. **Anchoring (Axis C)** can be developed alongside, but the *adaptive* grid depends on the scale field (LDSR), and the nerve depends on the anchor cover.
5. **Nerve integration** depends on PS, LDSR, and LCov; **capstone** depends on the nerve and LCov; **terminal** depends on all.

Gates that prune whole branches:

- **After LDS.** If the X-only scale already captures the structure (the y-aware variant adds little Truth-RMSE), keep X-only and the exact toolkit; do not pay the selection-aware df/band cost.
- **After LDR.** If the dimension field does not improve Truth-RMSE, it is not promoted; a global chart dimension is kept and LDSR reduces to scale-only.
- **After LDSR.** If co-regularized scale does not beat scale-frozen LDR at matched df, the program stays on LDR (fixed scale).
- **After the nerve.** If 2-simplex synchronization does not help, stay on the 1-skeleton.
- **After the capstone on synthetic.** If geometry-first basins are not more stable than the existing factorization, the capstone falls back to the COT dictionary route.

14 Immediate next actions (this revision)

1. **Land the Tier 0–4 integration.** dgp and t4 (uncertainty/bands) are merged; e19 (bandwidth/CV) and t2 (binary hygiene + the re-opened E0.6) are in their final audits and merges. This closes the validated LPS base and its toolkit.
2. **Decide the ladder sequencing.** Build **LDS** (the joint dimension/scale estimator) next, choosing the X-only-default / y-aware-opt-in fork; then choose whether LDR and LDSR land

as one joint field (E1.11 absorbs scale) or staged (LDR, then LDSR/E1.12). Spec in hand: the per-point-scale sketch.

3. **In parallel**, keep the synthetic enabler ahead: add the varying-scale generator so the ladder's gates have known-truth inputs.

Once the base is fully landed, the two refinement axes proceed in parallel, LCov inherits the ladder, the anchoring choice is exercised, and the nerve integrates them.

References

[1] *LPS Correctness and Validation: A Precise Experimental Plan* (the LPS base / Tier 0–4 spec), `project_briefs/lps_experimental_plan_2026-06-09`.

[2] *LPS/PS-LPS JASA-style audit memo* (PS-LPS linearity, df, λ_{sync} selection), `project_briefs/lps_ps_lps_jasa_style_audit_2026-06-09`.

[3] *LPS Local Dimension Regularization: Implementation Notes* (the LDR rung; the synthetic generators), `project_briefs/lps_local_dimension_regularization_implementation_notes_2026-06-10`.

[4] *Per-point / co-regularized chart scale for LPS* (the LDS/LDSR sketch; the X-only vs y-aware fork), `project_briefs/lps_per_point_scale_sketch_2026-06-13`.

[5] *Chart-Aware Local Association Structure on the L^∞ Atlas* (the LCov base), `project_briefs/chart_aware_local_association_design_2026-06-09`.

[6] *The Local-Chart Simplicial Complex as an Organizing Substrate for LPS* (the nerve integration), `project_briefs/lps_chart_nerve_substrate_2026-06-10`.

[7] *Occupancy Supports as Basins on the Chart Cover* (the capstone), `vaginal_community_trajectory_types/docs/occupancy_basins_on_chart_cover_bridge_2026-06-10`.

[8] *Worker–Auditor Research Project Workflow* (the audit-independence governance), `~/.codex/notes/workflows/worker_auditor_workflow`.